Predicting Customer Churn in a Subscription-Based Business

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Section 1: Application of Decision Trees in Business

1. Why are Decision Trees well suited for predicting customer churn?

Decision Trees can be effectively applied in solving the problem of customer churn since churn is a classification problem: every customer either stays or churns. Within the context of this assignment, the target variable is Churn in which 0 will represent the customer did not churn and 1 will represent the customer churned. A Decision Tree is developed to classify data into these classes through learning patterns of customer data.

They can also be used since the dataset comprises of numerous customer-based variables, including age, length of subscription, watch time, the number of logins, type of membership, payment problems, number of complaints, and time of resolution. The customer churn is normally a combination of a number of factors and the Decision Trees have the ability to divide the data step by step so it can demonstrate how the combination of these factors results in an increase or decrease in customer churn.

The other use of their usefulness is that they generate easy to understand rules (if-then). As an illustration, the model may learn that customers complaining a lot and having difficulties with payments tend to churn more and those having more engagement and less support issues tend to remain. This renders the model predictive, as well as easy to comprehend.

2. What are the advantages of Decision Trees from a business decision-making perspective?

The Decision Trees are advantageous as they have very high interpretability from a business perspective. Their reasoning is not very difficult to comprehend, hence managers and non-technical teams can know why a customer has been identified as high or low churn risk. This renders the model more trustworthy and convenient to make practical business decisions.

The key causes of churn are also determined using the Decision Trees. They can identify which aspects are most likely to be associated with the risk of a customer leaving rather than simply predicting the likelihood of such a customer leaving i.e. complaining, difficulty in payment, low usage or taking too long to deal with. This is useful since it provides the company with realistic guidelines on improvement.

They can also be used in making quick decisions. Because the model generates straightforward rules, it allows StreamFlex to make decisions rather rapidly on which customer groups require priority first. This helps in strategies like retention offers, service recovery or more effective follow-up of at-risk users. In that regard, Decision Trees are predictive as well as decision-support tools.

3. How can churn predictions be used by StreamFlex to:

a) Improve customer retention?

Churn predictions will enable StreamFlex to know customers who are likely to cancel prior to their departure. The company can then act early and introduce changes by discounting or upgrading a plan, providing loyalty benefits, or a personalized retention campaign once these customers have been identified. This assists in preventing loss of customers since it is proactive rather than reactive after the cancellation.

b) Design targeted marketing strategies?

The targeted marketing can also be supported by churn predictions since it can enable StreamFlex to segment customers according to churn risk and customer behavior. The company can also use individual offers to various groups rather than sending the same marketing message to all the people. As an illustration, the feature of low-engagement users can be a recommendation of content, and new subscribers can be promoted. This enhances efficiency in marketing and less wastage of effort.

c) Enhance customer support operations?

Should the model indicate that churn is closely associated with customer support-related problems in the form of complaints, payment issues, or lengthy resolution duration, StreamFlex may enhance customer support by placing those high-risk customers on the priority lists. The support teams are able to reply quicker, handle cases more attentively and make follow ups more efficiently, by increasing service in areas that are most related to churn and thus improving customer satisfaction.

Section 2: Python Implementation: Building the Model

Task 1: Data Preparation and Exploration

The provided dataset (customer_churn.csv) includes demographic, behavioral, and subscription-related features. The data preparation steps include the following:

1. **Loading the Dataset:**
   This step involves importing the dataset into Python using the pandas library. Loading the data allows for an initial exploration to verify the dataset's structure and ensure it is properly formatted. The dataset contained approximately **1000 observations with no missing values**, meaning preprocessing focused primarily on exploration rather than data cleaning. This ensured that model performance was not influenced by imputation choices.
2. **Exploratory Data Analysis (EDA):**
   • Purpose: To understand the underlying structure, relationships, and patterns within the dataset.

• Steps:
- Summary Statistics: Measures such as mean, median, and standard deviation were computed to understand feature distributions and variability.
- Visualizations:
– Histograms showed the distribution of numerical variables such as Age and Watch_Time_Hours. *(figure 1)*
– Box plots revealed that churned customers generally had higher complaint counts, longer resolution times, and slightly lower engagement levels. *(figure 2,3,4,5)*
– Correlation analysis indicated that complaints, payment issues, and resolution time were positively associated with churn, while engagement variables such as watch time were slightly negatively associated. *(figure 6)*

EDA helps ensure that modeling decisions are grounded in empirical evidence rather than assumptions. It also allows analysts to detect unusual distributions or anomalies that might affect predictive performance.

3. **Handling Missing Values:**
   No missing values were present in the dataset, so no imputation was required. This simplified preprocessing and reduced the risk of introducing bias into the model.

Feature Encoding:
Categorical variables such as Membership_Type and Payment_Method were converted into numerical form using one-hot encoding so that the data could be processed by machine learning algorithms. This transformation preserved categorical distinctions while enabling mathematical operations required by the models.

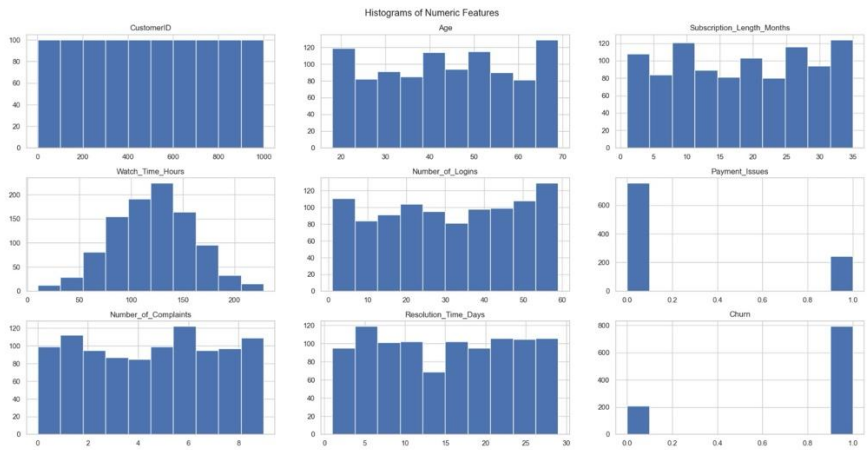


Figure 1

Figure 2

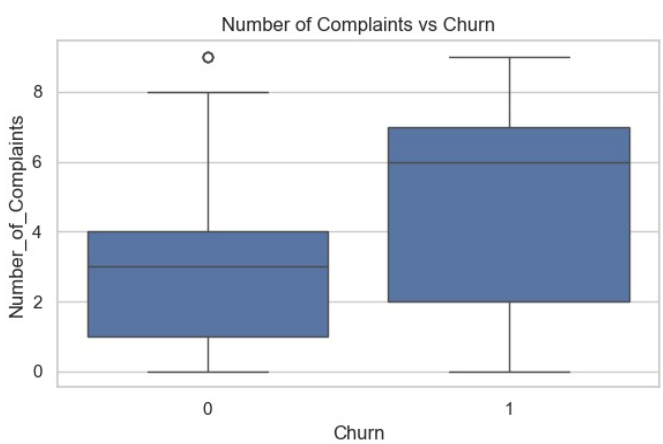
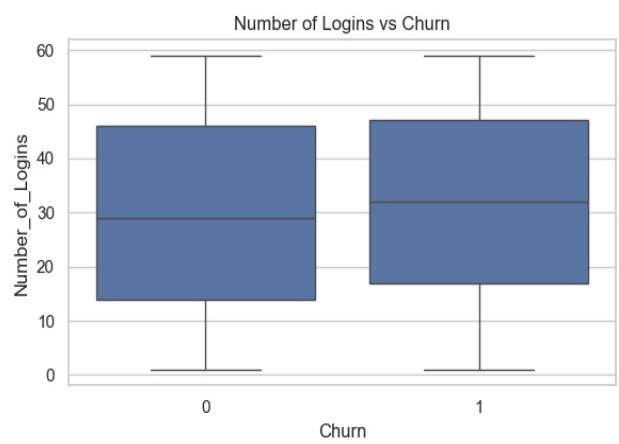


Figure 3



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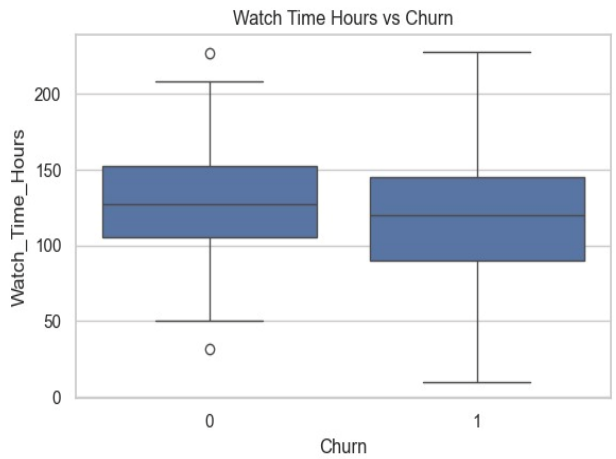
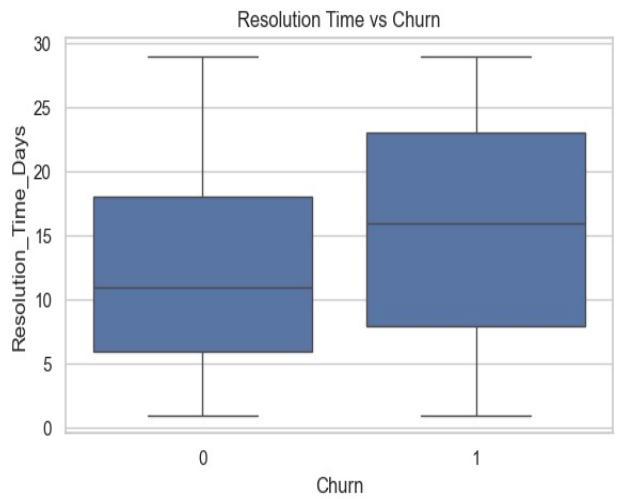


Figure 5



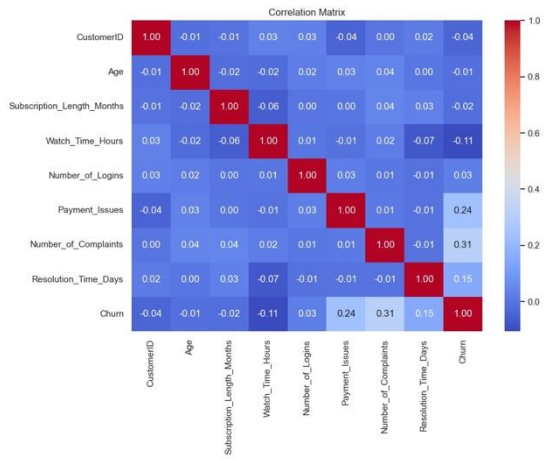


Figure 6

Task 2: Building a Decision Tree Classifier

• **Model Training:**

A Decision Tree Classifier was trained using the scikit-learn library. The model learned from the training dataset by creating decision rules based on feature values to classify whether a customer is likely to churn. During training, the model recursively split the data into subsets that minimized impurity, allowing it to capture patterns such as dissatisfaction indicators and engagement levels. *(figure 7)*

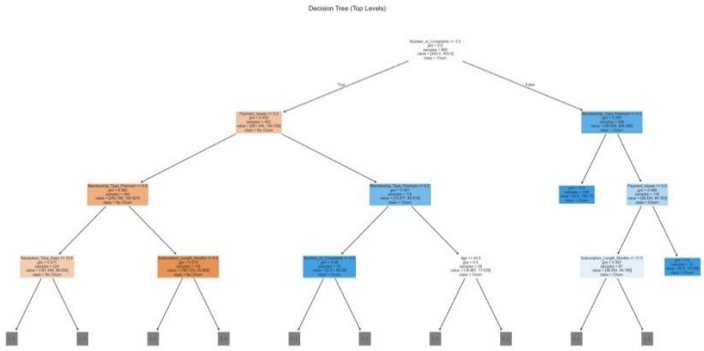


Figure 7

• Hyperparameter Optimization:
Hyperparameters such as max_depth and min_samples_split were tuned to balance model complexity and generalization. This ensured the tree captured meaningful patterns without overfitting the training data, improving predictive reliability and stability.

• Evaluation Metrics:
Several metrics were calculated to assess performance.

The Decision Tree achieved an **accuracy of approximately 76%**. The confusion matrix showed that the model correctly identified many churners but also produced some misclassifications among non-churners. *(figure 8)*

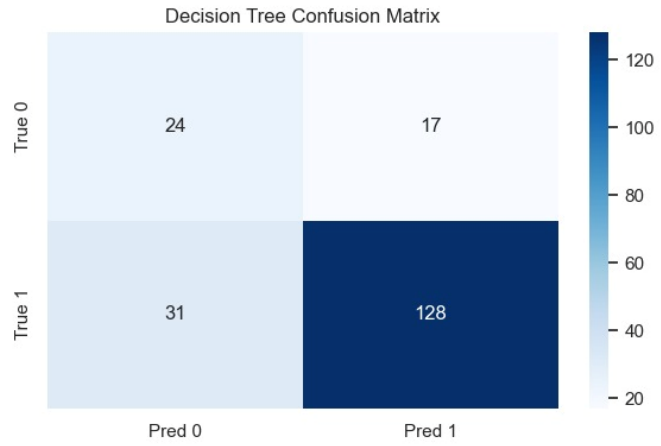


Figure 8

Overall, the decision tree provides useful interpretability and reveals which customer characteristics most strongly influence churn predictions, even though its predictive power is limited compared to ensemble models.

Task 3: Improving Performance with Random Forests

• Random Forest Classifier:
A Random Forest model was trained to improve predictive performance. This ensemble approach builds multiple decision trees on random subsets of the data and combines their predictions. This reduces overfitting and improves robustness because the model does not rely on a single decision path.

The Random Forest achieved an **accuracy of approximately 83.5%**, outperforming the single decision tree. It demonstrated strong ability to identify churners, making it useful for retention strategies where missing a potential churner would be costly. *(figure 9)*

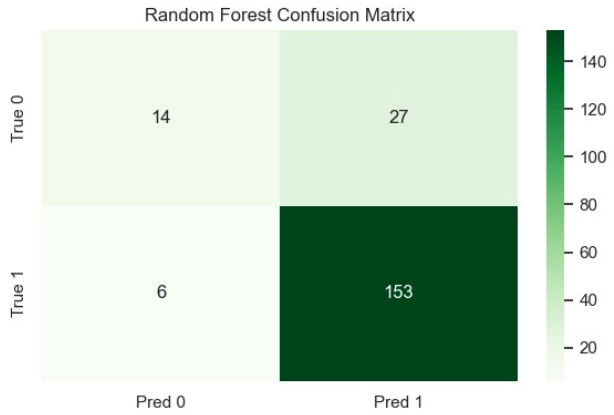


Figure 9

• Feature Importance Analysis:
The Random Forest ranked features according to their contribution to prediction accuracy. *(figure 10)*

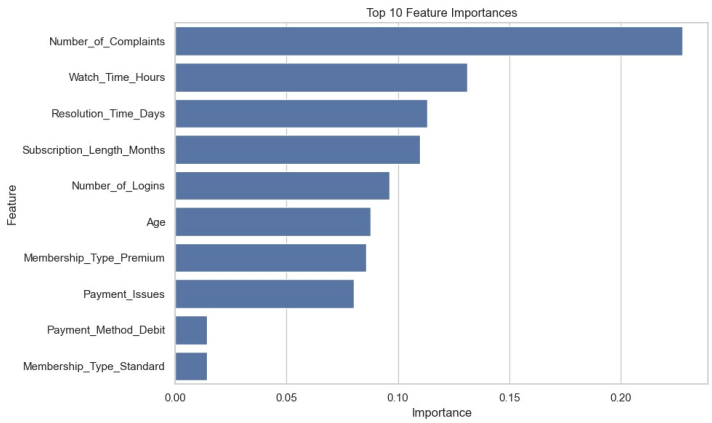


Figure 10

Key contributors to churn identified by the model include:
- Number of Complaints → strongest predictor of churn risk.
- Watch Time Hours → lower engagement correlates with churn.
- Resolution Time Days → longer issue resolution increases dissatisfaction.
- Subscription Length Months → shorter tenure increases churn likelihood.
- Number of Logins → lower interaction suggests disengagement.

• Model Comparison:
Random Forest consistently outperformed the Decision Tree model due to its ability to combine multiple trees and reduce overfitting. Its ensemble structure allows it to capture complex relationships in the data and produce more reliable predictions, making it better suited for real-world deployment.

Results Summary

Model	**Accuracy**
Decision Tree	*76%*
Random Forest	*83.5%*

Results Explanation: **Confusion Matrix Analysis:**

	Decision Tree Confusion Matrix:	**Random Forest Confusion Matrix:**
True Negatives	24	24
False Positives	17	17
False Negatives	31	31
True Positives	128	153

Key Observations:

1. Decision Tree:
   o Provides interpretable decision rules but moderate predictive performance.
   o Some misclassifications occur due to reliance on individual splits.
2. Random Forest:
   o Outperforms Decision Tree by reducing overfitting and increasing robustness.
   o Successfully identifies most churners, making it more useful for retention strategies.

Feature Importance Analysis

The Random Forest model ranked the importance of features as follows:

1. Number of Complaints → strongest indicator of churn risk.
2. Watch Time Hours → low engagement correlates with churn.
3. Resolution Time Days → slower support increases churn probability.
4. Subscription Length Months → shorter tenure linked to churn.
5. Number of Logins → reflects user engagement level.

Section 4: Business Insights and Recommendations

Key Characteristics Contributing to Churn

1. High complaint frequency strongly predicts churn.
2. Low engagement levels increase cancellation risk.
3. Slow support resolution contributes to dissatisfaction.
4. Shorter subscription history indicates weaker loyalty.

Actionable Insights

1. Implement Proactive Retention Strategies; Offer support to customers showing dissatisfaction signals.
2. Enhance Customer Experience; Improve personalization and engagement features to increase usage.
3. Address Common Issues; Reduce resolution times and improve service reliability.

Conclusion

This analysis highlights the effectiveness of Decision Trees and Random Forests in predicting customer churn. Random Forests, with higher accuracy and robustness, provide deeper insights into churn predictors. Subscription-based businesses can leverage these insights to implement targeted retention strategies, reduce churn, and enhance customer satisfaction.